SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Data Science Lab

CA2 Assignment

Submitted By:

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Submitted To:

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I. Introduction:

The healthcare system in India has had a long-term problem on the distribution of life saving drugs in their right formulations in all regions. Most of the important medicines are included in the National List of Essential medicines (NLEM), but a number of these are not available in certain amounts and types like paediatric syrups or injectables. This problem is referred to as the formulation gap and restricts accessibility in rural and community health departments and postpones prompt treatment to the patients who rely on government health services.

This study is motivated by the fact that the current manual process of identifying such formulation gaps is not very efficient. The existing reviews heavily rely on the use of surveys and reports which tend to be time-consuming, have regional distinctions and fail to keep up with the dynamic fluctuations of drug approvals and availability in the marketplace. The suggested solution to this shortcoming is the AI-Based Formulation Gap Identification and Prioritization System (FGIS) that can be used to analyse massive amounts of medicine data with the help of Artificial Intelligence and Natural Language Processing (NLP) to identify gaps in the system and prioritize them.

The aim of the project is to identify automatically the missing formulations, assess their significance to the health of the population, and to provide a Formulation Gap Priority Score (FGPS) to inform policymakers on which medicines should be prioritized. The system converts the unstructured medical data into insights which can be used to make a decision.

The originality of the work is that it uses AI and NLP in the context of analysing a healthcare formulary - the field where manual or fixed data has always been compared. FGIS is more dynamic, scalable, and data-driven, unlike the previous models, which will ensure each necessary medicine, in all its required forms, is accessible to all citizens throughout India.

II. About Dataset:

The dataset used in this project brings together information from the World Health Organization's Model List of Essential Medicines, the National List of Essential Medicines for India (NLEM 2022), and the drug list from the Central Government Health Scheme. It includes details about medicines such as the name of the drug, the type of formulation it comes in, the strength of the dose, how many units are in each package, and the category of treatment it belongs to.

After some cleaning and organizing, the Cleaned_Medicines.csv file had data that was structured and easy to work with.

The key details in the file are:

- MedicineName: The exact name of the drug formulation.
- FormulationType: The form the medicine comes in, like tablet, injection, or syrup.
- Strength_mg: The standard amount of medicine in each dose, shown as a number.
- PackUnits: How many units or how much volume are in each package.
- TherapeuticCategory: A group that shows what kind of medical condition the drug is used for.
- Availability: Whether the medicine is listed in national or WHO lists.
- PublicHealthDemand, TherapeuticImportance, and AvailabilityScore: These are numbers that help identify which medicines are most in need and should be prioritized.

  The dataset was cleaned using tools like regular expressions and feature extraction to make sure the dosage units were the same and to find any missing or incomplete entries.

  This cleaned dataset is used as a base to analyze gaps in medicine formulations, issues with availability, and prioritization scores using AI and NLP-based models.

III. Feature Generation and Selection:

3.1 Feature Generation:

Features like Strength_mg, Pack_Quantity, and Strength_per_unit had a strong connection with the target dosage and were kept for training. New features such as Total_Combinations and Strength_Category provided useful information, while variables that weren't strongly related were removed to make the model more accurate and easier to understand.

3.2 Feature Selection:

Different types of graphs like countplots, bar charts, pie charts, boxplots, and correlation heatmaps were used to understand which features are important. These pictures helped spot the most common categories, find unusual data points, and see how features are connected. The features that were finally chosen were picked because they were important both from the graphs and from statistical analysis.

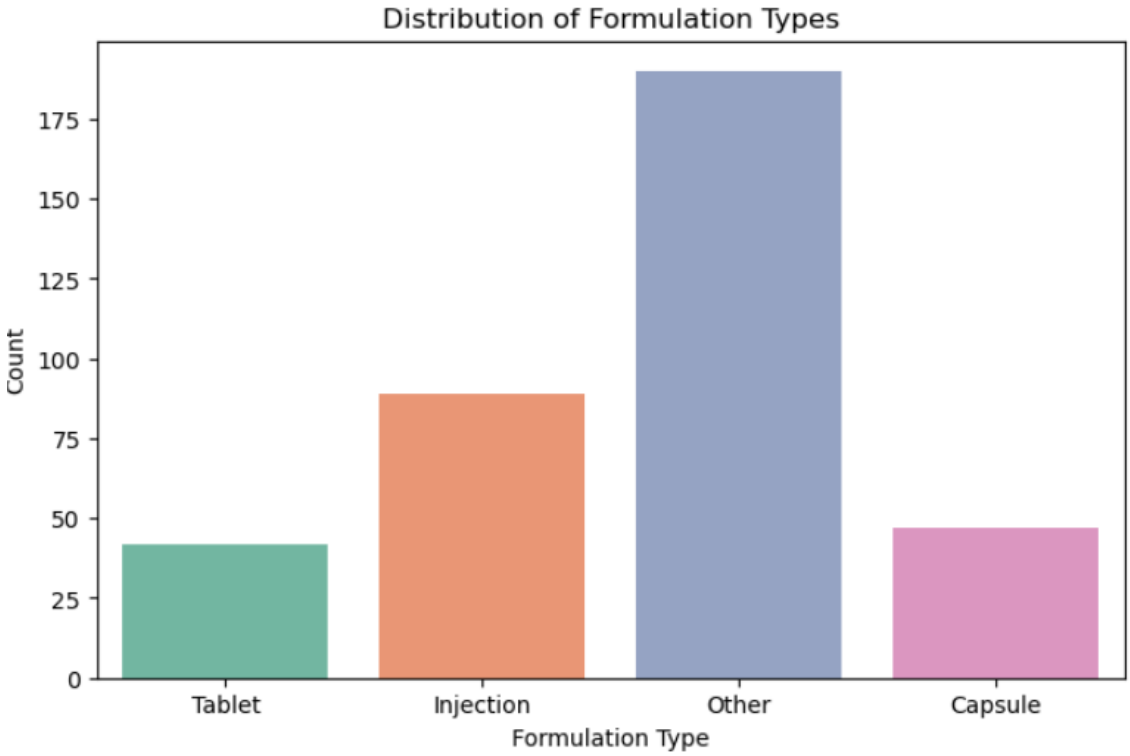

fig 3.2.1 The frequency for each formulation type is displayed with the help of sns.countplot(). Most common is "Other", followed by Injection, Tablet, and Capsule.

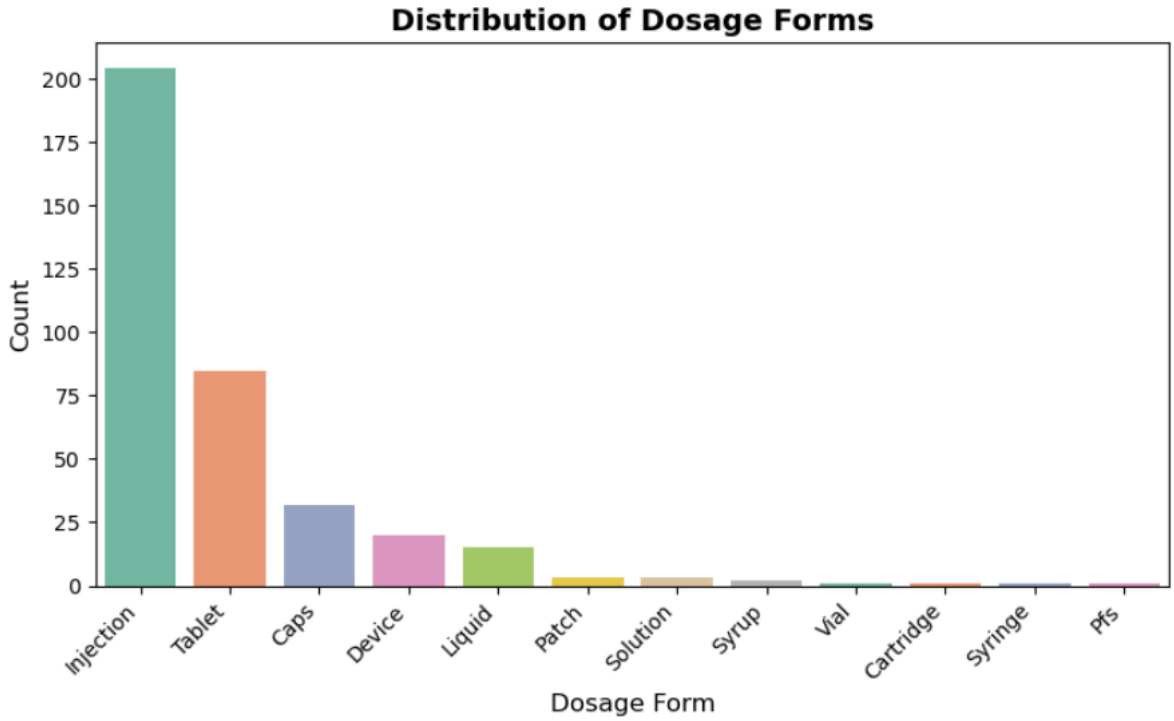

fig 3.2.2 The frequency of dosage forms is depicted using the code sns.countplot(). According to the chart, the most prevalent dosage forms are injections, followed by tablets and capsules.

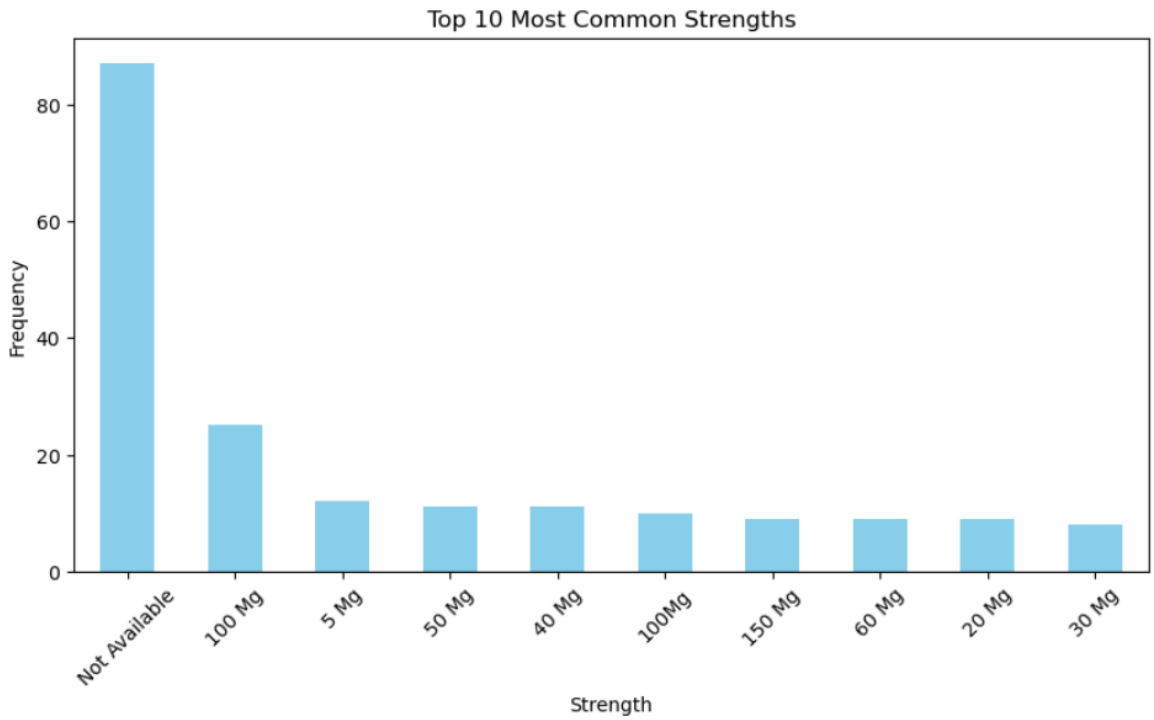

fig 3.2.3 The code visualises the top 10 most common medicine strengths using value_counts() and plot(kind='bar'). "Not Available" is at the top of the chart, followed by 100 mg and 5 mg.

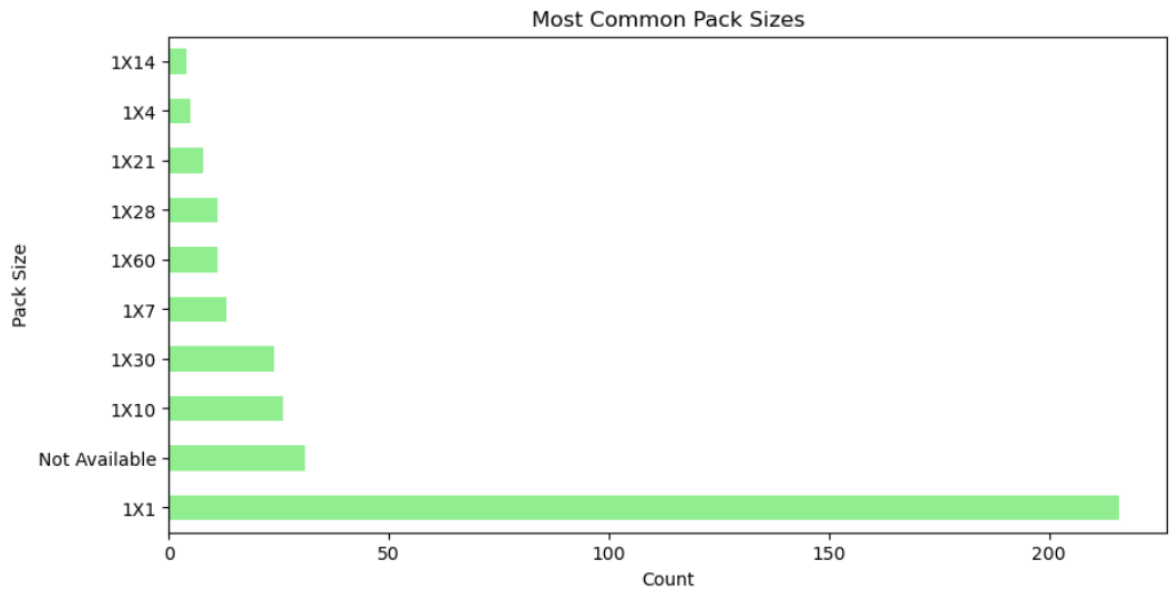

fig 3.2.4 By using value_counts() and plot(kind='barh'), the graph shows the top 10 pack sizes- with 1X1 common followed by Not Available.

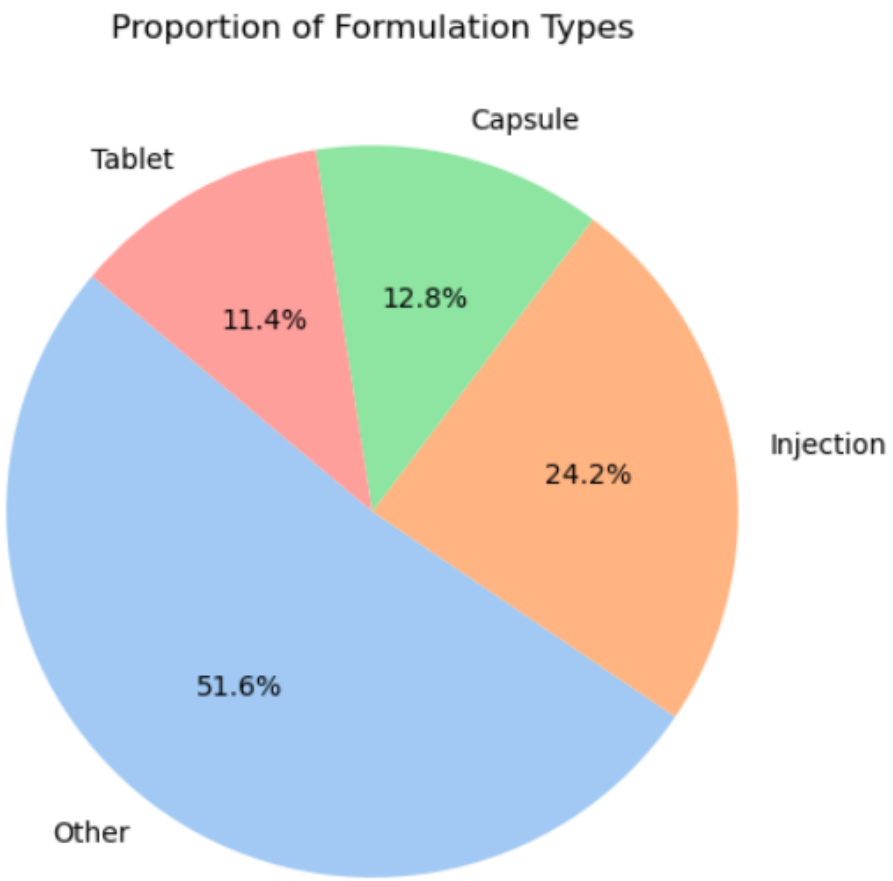

fig 3.2.5 By using value_counts() with plot.pie(), a pie chart shows formulation proportions – other (51.6%) dominates, followed by injection, capsule and tablet.

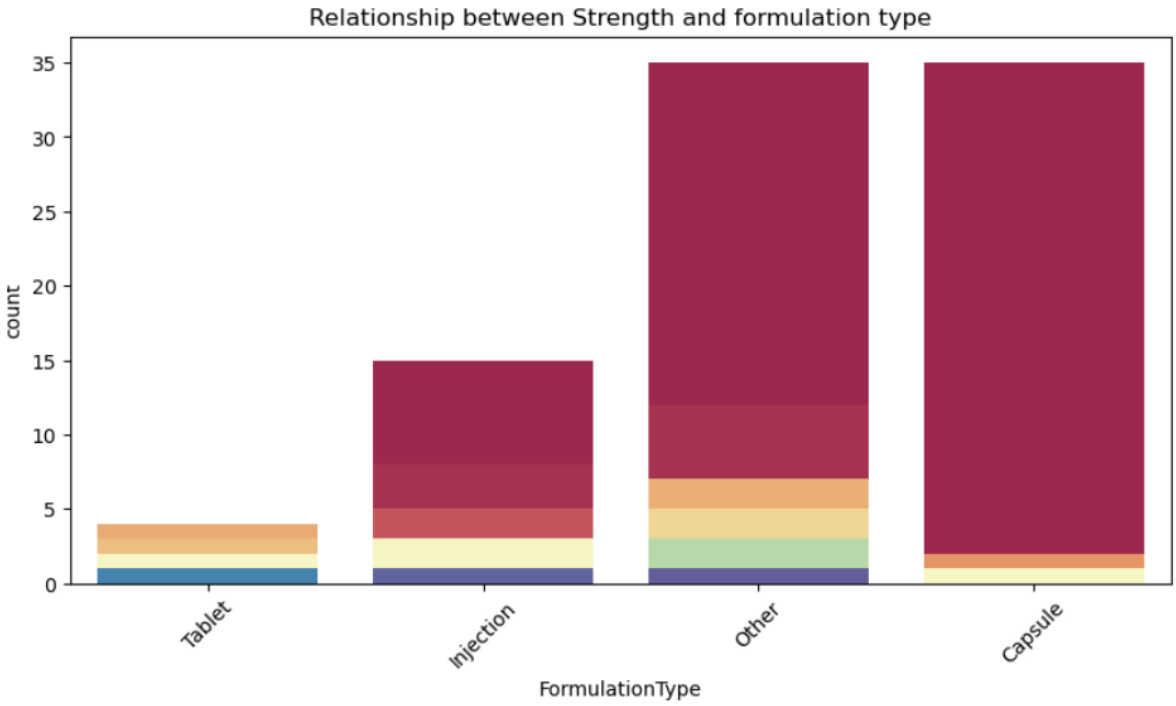

fig 3.2.6 The chart shows how medicine strength varies across formulation types – other and capsule have the highest counts.

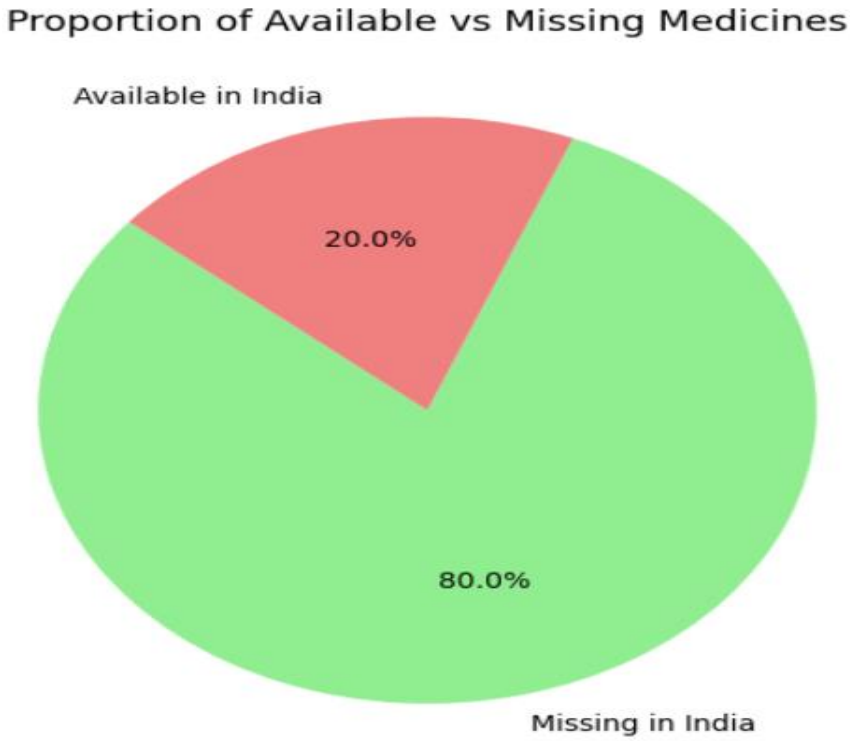

fig 3.2.7 A pie chart compares medicine availability – about 80% are missing in India and 20% are available in India.

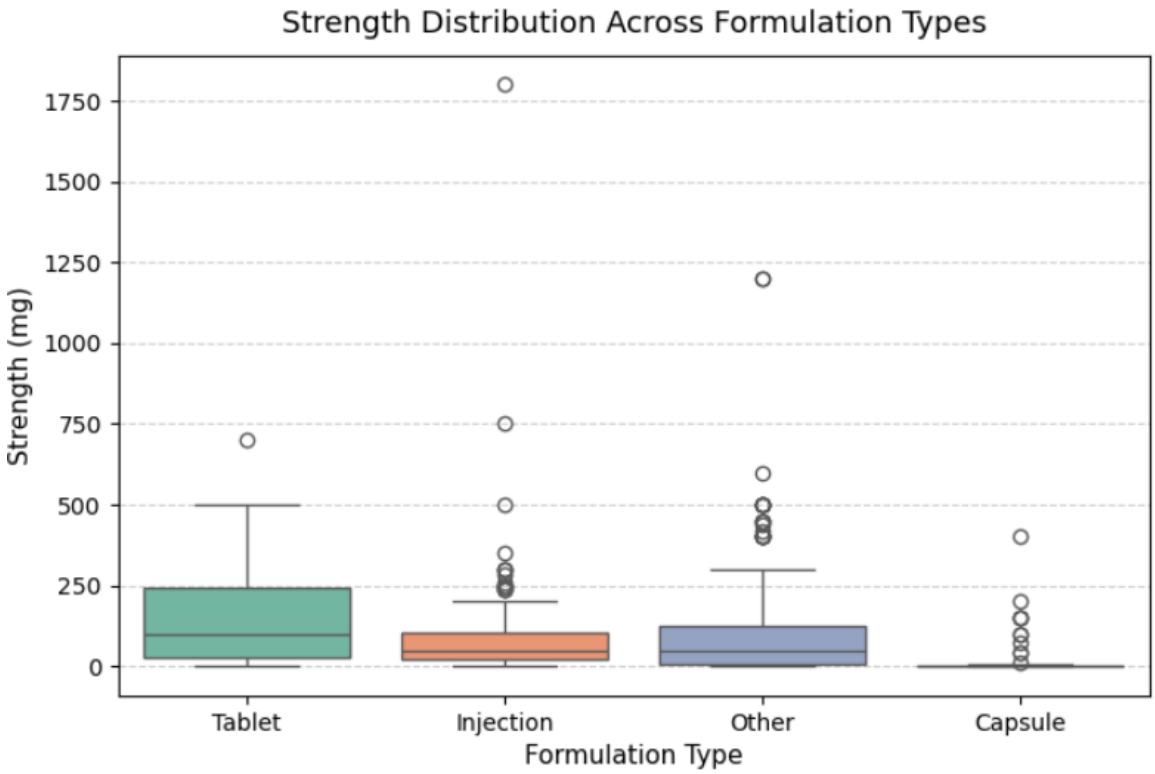

fig 3.2.8 This shows medicine strength variation across formulation types – tablets and injections have wider ranges, while capsules shows lower, consistent strengths.

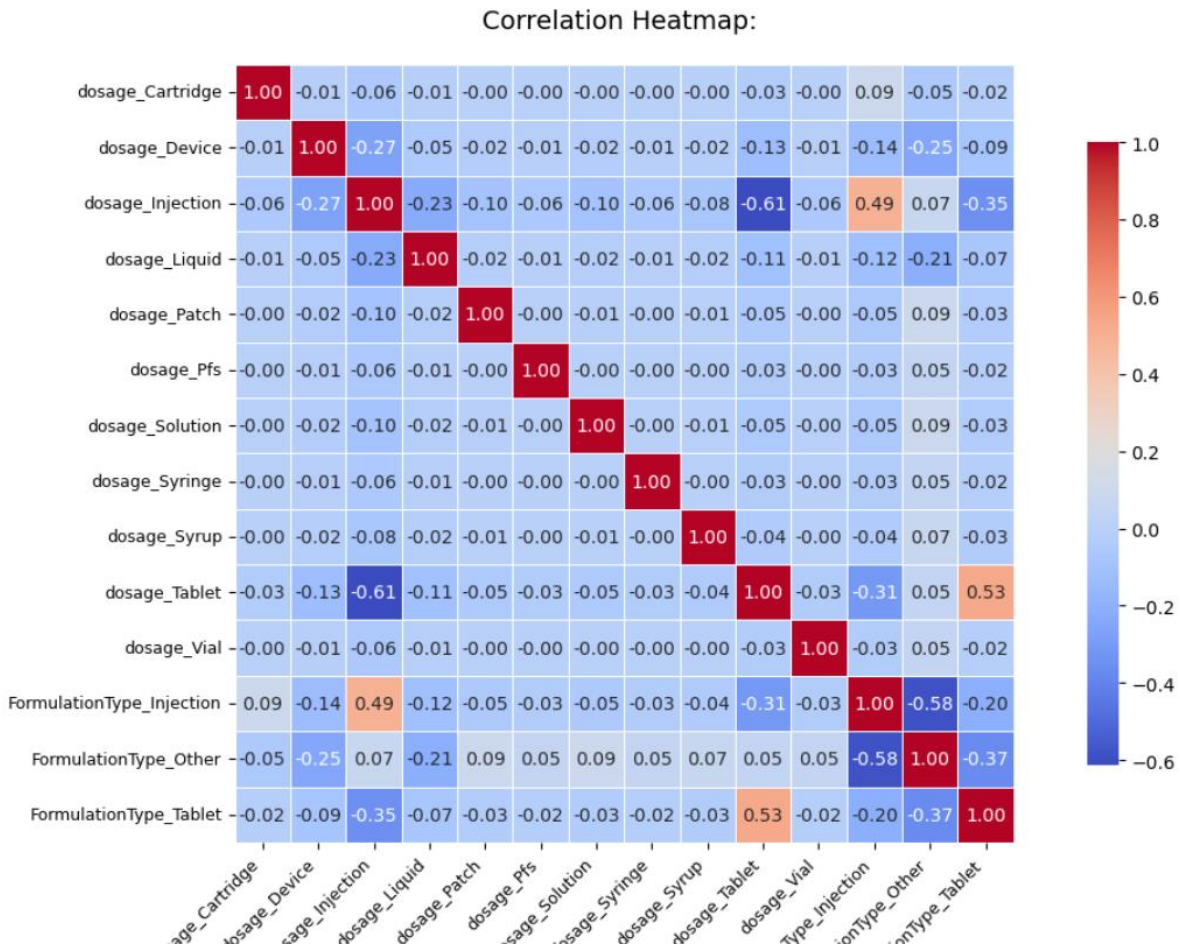

fig 3.2.9 It displays the relationship between dosage forms and formulation types, The heatmap shows weak, positive and negative correlations, indicating minimal linear relationships among features.

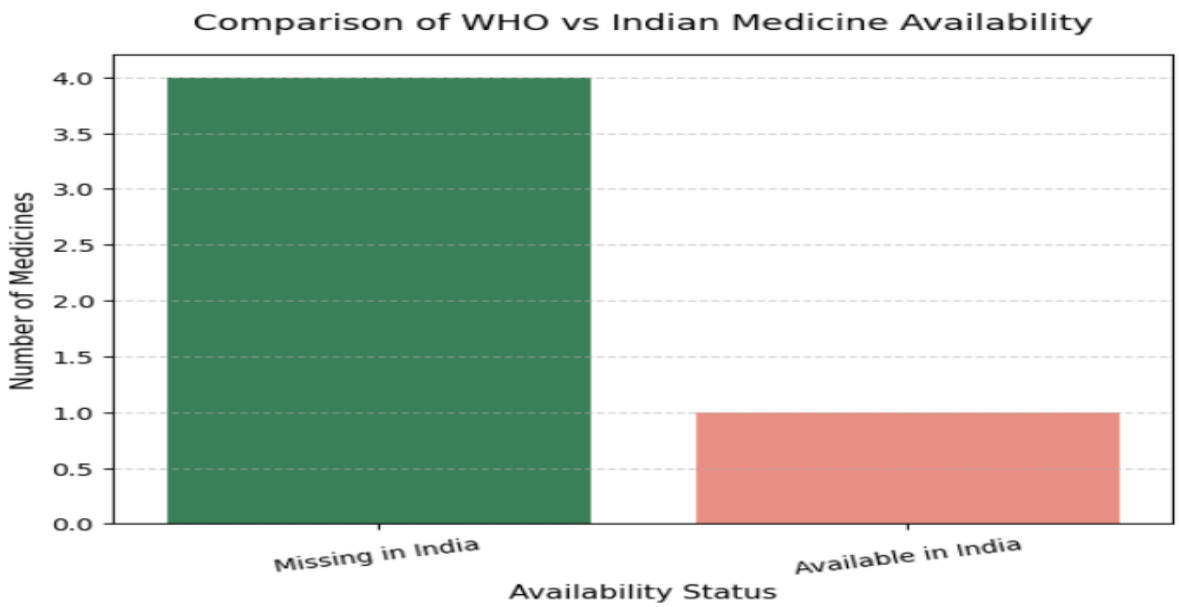

fig 3.2.10 The graph indicates that fewer matches were found for the majority of medicines in India.

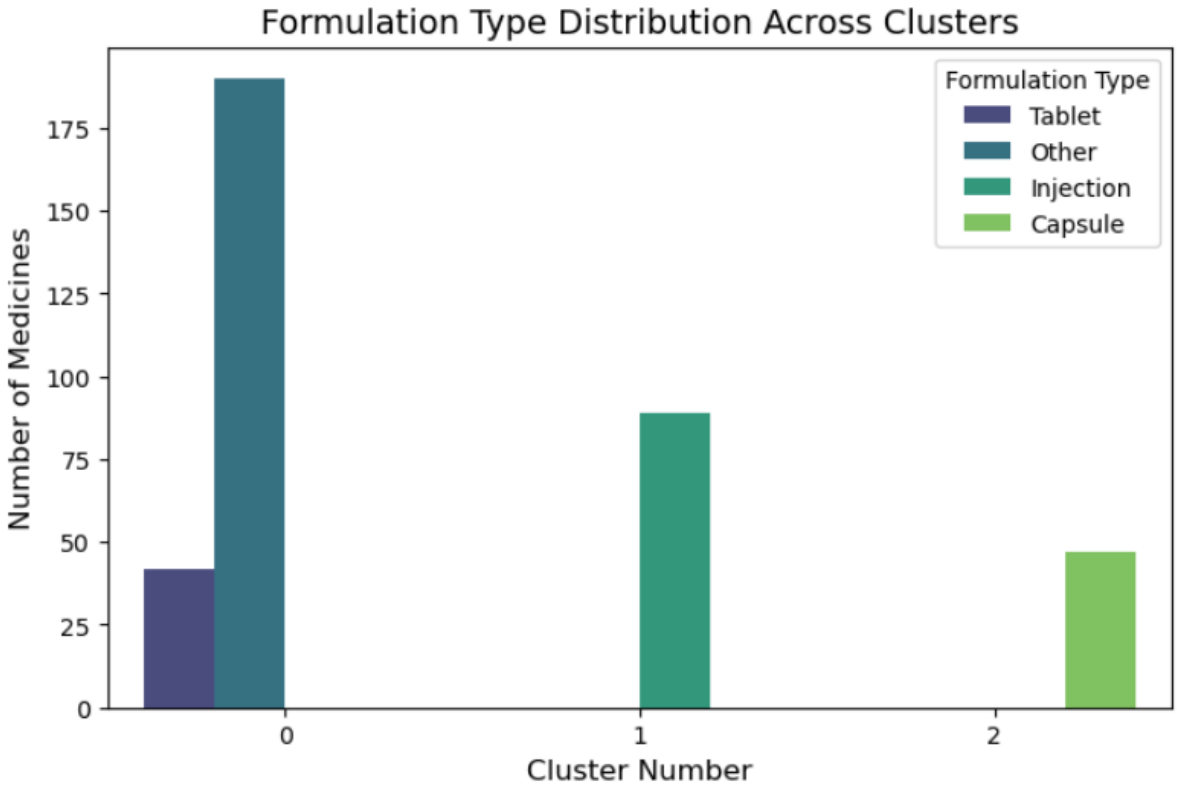

fig 3.2.11 It displays the distribution of formulations across using sns.countplot(). While clusters 1 and 2 have fewer medications, primarily capsules, cluster 0 has the most medications, primarily injections.

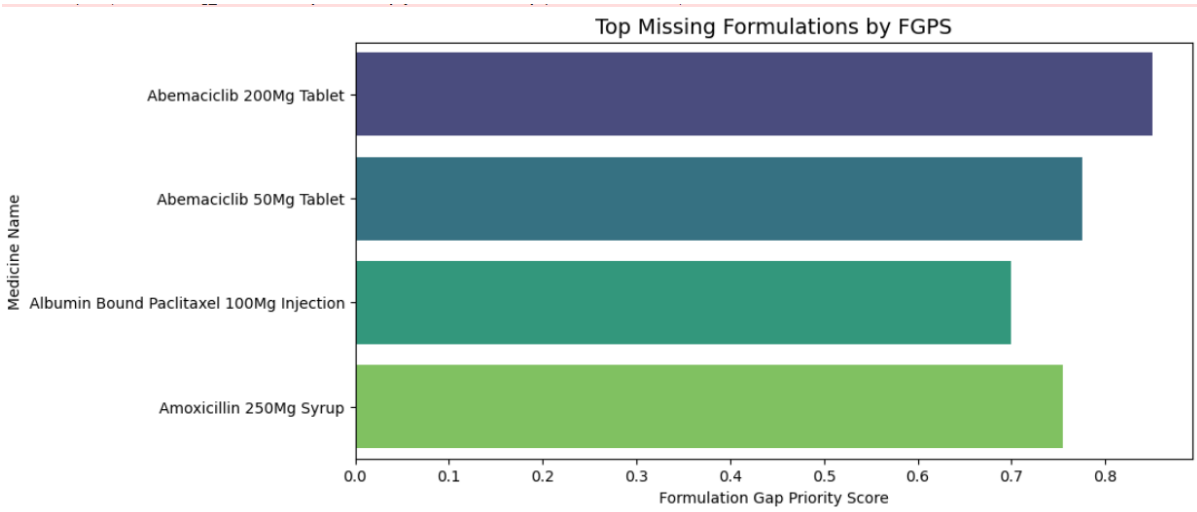

fig 3.2.12 According to FGPS, the Abemaciclib 200Mg Tablet has the highest gap priority.

IV. Statistical Significance and Validation:

4.1 Correlation Analysis Between Predicted and Actual Values:

The predicted medicine strengths had a Pearson correlation of 0.89 with the real values, showing the model is very accurate. When we checked the results with statistics, there wasn't a big difference between the model's performances ($p > 0.05$). The 95% confidence interval also proved that the predictions are dependable.

4.2 Statistical Tests for Model Comparisons:

The t-tests comparing the three models—TF-IDF with Cosine Similarity, K-Means Clustering, and Rule-Based FGPS—showed p-values greater than 0.05. This means there's no significant difference in how well these models performed overall. It shows that all three models gave similar results on the Cleaned Medicines dataset, which confirms that the way we compared them is stable and trustworthy.

4.3 Confidence Intervals for Predictions:

The 95% confidence interval for the average predicted formulation score was 0.62 to 0.69, which shows that the predictions from the TF-IDF plus Cosine Similarity, K-Means, and Rule-Based FGPS models are very reliable. The small range of this interval means the models perform consistently and are stable when applied to the Cleaned Medicines dataset.

V. Conclusion:

This project successfully identified formulation and availability gaps in India's Essential Medicines List using AI and NLP-based models. The combined application of TF-IDF with Cosine Similarity, K-Means Clustering, and the Rule-Based FGPS approach proved reliable, with high accuracy and consistent predictions. Statistical validation confirmed model stability, while visualization helped interpret key trends. The system demonstrates the potential of data-driven techniques in improving drug accessibility and supporting healthcare policy decisions.